A DERIVED LIMIT OF A COMPATIBLE SYSTEM OF PSEUDO-COHERENT OBJECTS OVER SURJECTIVE RING MAPS IS PSEUDO-COHERENT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 28335-28351 (canonical tag 0CQF) and the counted proof is the proof environment at source lines 28353-28375, which immediately follows it and is its own proof.

Editorial note. The five references are four earlier results of this chapter plus a remark of this chapter cited alongside one of them; the kinds recorded are the type words appearing in the labels themselves, following the lane convention, and none of them is the name of the environment in which the label lives.

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Allowed standard prerequisites (editorial).

- A pseudo-coherent object over a ring is represented by a bounded above complex of finite free modules, and pseudo-coherence is preserved under a surjection with locally nilpotent kernel when the target object is pseudo-coherent. invoked at proof lines 28353-28359. Produces the representative over the first ring and propagates pseudo-coherence to every member of the system.
- Maps in the derived category between the members of the system may be lifted to maps of complexes of finite free modules inducing isomorphisms after reduction to the previous ring. invoked at proof lines 28359-28365. Builds the system of complexes whose limit represents the derived limit.
- The derived limit of a system of complexes is represented by the limit of the complexes, and the limit of finite free modules of bounded rank is again finite free of that rank. used at proof lines 28365-28374. Shows that the limit complex is a bounded above complex of finite free modules, which gives pseudo-coherence and the final assertion.

Source reference. Original definition 064Q: sources/stacks-more-algebra/pinned-original-source.tex, source lines 15874–15874. The reference supplies the definition used at proof lines 28353-28355 to choose the bounded above complex of finite free modules; the induction built on it is inside the counted span.

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Source reference. Original lemma 0H76: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19498–19498. The reference is applied at proof lines 28355-28357 to conclude that every object of the system is pseudo-coherent; the lifting argument that follows is inside the counted span.

Source reference. Original lemma 0BCB: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19517–19517. The reference is invoked at proof lines 28359-28361 to construct the complexes and maps of the system with strict reductions; the limit computation that uses them is inside the counted span.

Source reference. Original lemma 091D: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23574–23574. The reference identifies the derived limit with the limit of the system of complexes, used at proof lines 28365-28367; the reading off of finite freeness afterwards is inside the counted span.

Source reference. Original remark 07KZ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23753–23753. The remark is cited jointly with the previous reference at proof lines 28365-28367 and carries no separate mathematical input; it is a pointer accompanying that identification.

Lemma 0.2. *Let $A = \lim A_n$ be a limit of an inverse system (A_n) of rings. Suppose given $K_n \in D(A_n)$ and maps $K_{n+1} \to K_n$ in $D(A_{n+1})$. Assume*

1. *the transition maps $A_{n+1} \to A_n$ are surjective with locally nilpotent kernels,*
2. *either all K_n are pseudo-coherent or there exists an integer n_0 such that K_{n_0} is pseudo-coherent and the kernels of $A_{n+1} \to A_n$ are nilpotent ideals for $n \geq n_0$,*
3. *the maps induce isomorphisms $K_{n+1} \otimes^{\mathbf{L}}_{A_{n+1}} A_n \to K_n$.*

Then $K = R\lim K_n$ is a pseudo-coherent object of $D(A)$ and $K \otimes^{\mathbf{L}}_A A_n \to K_n$ is an isomorphism for all n.

Proof. By assumption we can find a bounded above complex of finite free A_1-modules $P_1^\bullet$ representing K_1, see Definition 064Q. By Lemma 0H76 we conclude that K_n is pseudo-coherent for all n. Then by Lemma 0BCB we can, by induction on $n > 1$, find complexes $P_n^\bullet$ of finite free A_n-modules representing K_n and maps $P_n^\bullet \to P_{n-1}^\bullet$ representing the maps $K_n \to K_{n-1}$ inducing isomorphisms (!) of complexes $P_n^\bullet \otimes_{A_n} A_{n-1} \to P_{n-1}^\bullet$. Thus $K = R\lim K_n$ is represented by $P^\bullet = \lim P_n^\bullet$, see Lemma 091D and Remark 07KZ. Since P_n^i is a finite free A_n-module for each n and $A = \lim A_n$ we see that P^i is finite free of the same rank as P_1^i for each i. This means that K is pseudo-coherent. It also follows that $K \otimes^{\mathbf{L}}_A A_n$ is represented by $P^\bullet \otimes_A A_n = P_n^\bullet$ which proves the final assertion. □